

# NVFP4-DiT: Efficient 4-Bit Audio-Guided Video Diffusion Transformers for Low-Cost Video Generation

Md Rakibul Islam Raihan, Peng Zhang

**Abstract**—Audio-guided video diffusion transformers produce high-quality visual content, but their training and inference costs remain a major barrier to practical deployment. We present NVFP4-DiT, a low-precision framework for audio-guided text-to-video generation in 4-bit floating-point precision using NVIDIA’s NVFP4 format. The framework combines a theoretical analysis of FP4 diffusion under block-wise scaling, including a sufficient condition for preserving temporal coherence, with an adaptive block-wise quantization strategy and quantization-aware training (QAT) using learnable scales for cross-modal attention. To support efficient video generation, we further describe custom Triton kernels for FP4-packed matrix multiplication and attention. Experiments on WebVid-10M, VGGSound, and UCF-101 evaluate generation quality, synchronization, latency, throughput, and memory use across multiple DiT variants. The results indicate that NVFP4-DiT preserves competitive video quality and audio-visual alignment while substantially reducing memory footprint and computational cost.

**Index Terms**—Diffusion Transformers, Quantization-Aware Training, Low-Precision Learning, NVFP4, Neural Network Compression, Multimodal Learning.

## I. INTRODUCTION

### A. Motivation

Diffusion transformers (DiTs) [?] have become a powerful foundation for modern video generation, enabling high-fidelity synthesis from text and audio prompts [?], [?], [?]. Recent systems such as Sora [?], Stable Video Diffusion [?], and ModelScope-T2V [?] demonstrate strong capability in producing temporally coherent and semantically aligned video content. However, the computational cost of these models remains prohibitive: training a single DiT can require hundreds to thousands of GPU days, while inference can still demand several seconds per generated frame on high-end accelerators [?], [?].

This cost barrier limits the use of generative video models in practical video systems, including interactive media tools, low-latency content creation, and resource-constrained deployment. Although model compression techniques such as pruning [?], distillation [?], and quantization [?] have shown promise in image generation, video diffusion introduces additional challenges: temporal coherence must be preserved across frames, audio and visual streams must remain synchronized, and spatiotemporal attention incurs high computational complexity.

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### B. Low-Precision Opportunities

NVIDIA’s NVFP4 format [?]<sup>1</sup>—a 4-bit floating-point representation with 1 sign bit, 2 exponent bits, and 1 mantissa bit—offers unprecedented compression for deep neural networks. When combined with tensor core acceleration on H100 and B200 GPUs, FP4 can theoretically achieve 4× memory reduction and 3× throughput improvement over FP16 [?]. However, three critical questions remain unanswered:

- 1) **Theoretical:** Can diffusion models—which are notoriously sensitive to numerical precision—converge and generate coherent videos under extreme 4-bit quantization?
- 2) **Algorithmic:** How should quantization be adapted for the spatiotemporal dynamics of video diffusion, where temporal consistency demands precise attention patterns across frames?
- 3) **Systems:** What kernel-level optimizations are necessary to realize the theoretical gains of FP4 for video generation workloads?

### C. Contributions

This paper addresses these questions through **NVFP4-DiT**, a framework for efficient 4-bit training and inference of audio-guided video diffusion transformers. Our contributions are:

- 1) **Theoretical Analysis (Section ??):** We derive convergence bounds for FP4-trained diffusion models under block-wise scaling and provide a sufficient condition for preserving temporal coherence, namely  $\|\Delta QK^T\|_F \leq \delta_t \cdot \|\mu\|_2$  for temporal difference  $\delta_t$  and mean attention  $\mu$ .
- 2) **Algorithmic Design (Section ??):** We introduce **adaptive block-wise quantization** for spatiotemporal diffusion layers and **quantization-aware training (QAT)** with learnable scale factors for cross-modal attention between audio spectrograms and visual latents.
- 3) **Systems Optimization (Section ??):** We develop custom Triton [?] kernels for FP4-packed matrix multiplication and attention, and we describe how the implementation integrates with vLLM [?] and TensorRT-LLM [?] for efficient batched inference.
- 4) **Empirical Evaluation (Section ??):** We evaluate NVFP4-DiT on WebVid-10M [?], VGGSound [?], and UCF-101 [?], showing competitive quality relative to BF16 together with substantial reductions in memory use, throughput cost, and training energy.

### D. Novelty and Scope

Prior low-precision and quantization efforts differ from this work in three concrete respects. First, existing FP4/INT4 studies target image diffusion or language/vision backbones [?], [?], [?] and learn quantization scales for static layers; none address the temporal-coherence and cross-modal (audio–visual) constraints that are unique to video generation. Second, while NVFP4 has been adopted for LLM training and inference [?], [?], its application to spatiotemporal diffusion transformers trained with quantization-aware learning has not been studied, and the learning-theoretic question of when 4-bit noise schedules converge while preserving temporal consistency remains open. Third, efficient video serving is typically treated at the systems level [?], [?] independently of the learned numerical format, whereas we co-design the low-precision representation with the serving kernels. NVFP4-DiT is, to our knowledge, the first framework that (i) derives convergence bounds for FP4 diffusion training under block-wise scaling, (ii) introduces learnable quantization scales for cross-modal attention via quantization-aware training, and (iii) delivers custom FP4 kernels for audio-guided video diffusion. In scope, the work aligns with the *IEEE Transactions on Neural Networks and Learning Systems* remit: learning algorithms for neural-network compression, low-precision training, and multimodal representation learning.

### E. Paper Organization

Section ?? reviews related work. Section ?? presents theoretical foundations. Section ?? describes our algorithmic framework. Section ?? details systems optimizations. Section ?? reports experimental results. Section ?? ablates design choices. Section ?? discusses limitations and impact. Section ?? concludes.

## II. RELATED WORK

### A. Video Diffusion Models

Diffusion models [?], [?] generate data by reversing a gradual noising process. For video, early approaches extended image diffusion with temporal layers [?], [?]. Recent transformer-based architectures [?], [?] unify spatial and temporal attention in a single DiT backbone, achieving state-of-the-art results. Audio-guided variants [?], [?] condition on mel-spectrograms via cross-attention, enabling synchronized audiovisual generation. However, all these models operate at FP16/BF16 precision, incurring substantial computational costs.

### B. Quantization for Generative Models

Quantization reduces precision to compress models and accelerate inference. Post-training quantization (PTQ) [?], [?] is applied after training but can degrade quality in generative models. Quantization-aware training (QAT) [?], [?] simulates quantization during training and is generally more robust at low bitwidths. Prior work on image diffusion has shown that low-precision deployment is feasible in less demanding settings [?]. For video, however, low-bit quantization remains

challenging because temporal coherence must be preserved across frames. Our work studies how 4-bit training can be stabilized in this setting.

### C. Low-Precision Formats

NVIDIA’s FP8 [?] and NVFP4 [?] formats enable extreme compression. FP8 (E4M3/E5M2) has been applied to LLM training [?] and inference [?], [?]. NVFP4—with 2 exponent bits and 1 mantissa bit—provides dynamic range up to  $2^{2^2} = 16$  and precision of  $2^{-1} = 0.5$  relative to scale. Prior work on FP4 has primarily focused on language and vision backbones [?], where the dominant concern is per-token accuracy rather than cross-frame consistency. More recent efforts extend NVFP4 to LLM training and inference [?], yet they leave two gaps that are central to this paper: (a) they do not consider generative video models, where a single quantized attention map must remain stable across hundreds of frames, and (b) they do not study cross-modal (audio–visual) attention, whose synchronization is highly sensitive to scale errors. Our work is the first to close both gaps for audio-guided video diffusion transformers.

### D. Systems for Efficient Inference

vLLM [?] introduced PagedAttention and continuous batching for LLM serving. TensorRT-LLM [?] provides optimized kernels and graph optimizations for transformer inference. In this work, we draw on design ideas from both systems and discuss how similar serving strategies can be adapted to spatiotemporal video diffusion workloads.

## III. THEORETICAL FOUNDATIONS

### A. Preliminaries

Let  $x_0 \in \mathbb{R}^{C \times F \times H \times W}$  represent a video tensor with  $C$  channels,  $F$  frames, height  $H$ , and width  $W$ . The forward diffusion process adds Gaussian noise over  $T$  timesteps:

$$q(x_t|x_{t-1}) = \mathcal{N}(x_t; \sqrt{1 - \beta_t}x_{t-1}, \beta_t I) \quad (1)$$

where  $\beta_t \in (0, 1)$  is a variance schedule. The reverse process learns a denoising network  $\epsilon_\theta(x_t, t, c)$  conditioned on text  $c_{\text{text}}$  and audio  $c_{\text{audio}}$ :

$$p_\theta(x_{t-1}|x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t, c), \sigma_t^2 I) \quad (2)$$

### B. FP4 Quantization Error Bounds

**Definition III.1** (NVFP4 Quantization). For a floating-point value  $v$  with scale factor  $s \in \mathbb{R}^+$ , the NVFP4 quantized value is:

$$Q_{\text{FP4}}(v; s) = s \cdot \text{clip}(\lfloor v/s \rfloor_{\text{FP4}}, -6, 6) \quad (3)$$

where  $\lfloor \cdot \rfloor_{\text{FP4}}$  rounds to the nearest representable FP4 value, and clipping limits the exponent range to  $[-6, 6]$  in the scaled domain.

**Lemma III.2** (Quantization Error Bound). For any  $v \in \mathbb{R}$ , the relative quantization error satisfies:

$$\left| \frac{Q_{FP4}(v; s) - v}{v} \right| \leq \epsilon_{FP4} = 2^{-3} = 0.125 \quad (4)$$

*Proof.* FP4 has 1 mantissa bit, giving a relative precision of  $2^{-1} = 0.5$  in the representable range. With optimal scaling, the maximum relative error is half of this, yielding 0.25. However, block-wise scaling with  $2 \times 2$  blocks reduces error to 0.125. The complete proof follows from [?, Theorem 1].  $\square$

**Theorem III.3** (Convergence of FP4-Diffusion). *Let  $\epsilon_{FP}$  be the denoising network trained with FP4 quantization. Under Assumptions 1-3 (Lipschitz continuity of  $\epsilon_\theta$ , bounded gradients, and smooth noise schedule), the reverse process error satisfies:*

$$\mathbb{E}[\|x_0 - \hat{x}_0\|_2^2] \leq \frac{C}{T} \sum_{t=1}^T (\epsilon_{FP4}^2 \cdot \mathbb{E}[\|\nabla_{x_t} \log p(x_t)\|_2^2] + \sigma_t^2) \quad (5)$$

where  $C$  is a constant depending on model architecture and  $T$  is the number of diffusion steps.

*Proof Sketch.* The proof extends the convergence analysis of DDIM [?] to account for quantization error. At each step, the FP4-quantized prediction introduces an error term bounded by  $\epsilon_{FP4} \|\nabla \log p(x_t)\|$ . Accumulating this term over  $T$  steps and applying the Lipschitz property yields the stated bound. A complete derivation is provided in Appendix A.  $\square$

**Corollary III.4.** *For sufficiently large  $T$  ( $T \geq 1000$ ), the FP4 convergence error approaches the FP16 baseline up to a multiplicative factor of  $(1 + O(\epsilon_{FP4}))$ .*

### C. Temporal Coherence Under Quantization

**Definition III.5** (Temporal Coherence). A generated video sequence  $\{x^{(f)}\}_{f=1}^F$  exhibits temporal coherence if the frame-to-frame difference satisfies:

$$\|x^{(f+1)} - x^{(f)}\|_F \leq \tau \cdot \|x^{(f)}\|_F \quad \forall f \quad (6)$$

for some  $\tau \ll 1$ .

**Theorem III.6** (Coherence Preservation Condition). *Let  $A = QK^T$  be the attention matrix in a DiT block. Under FP4 quantization, the temporal coherence error  $\Delta_{temp}$  satisfies:*

$$\Delta_{temp} \leq \frac{2\epsilon_{FP4}}{1 - \epsilon_{FP4}} \cdot \|\mu\|_2 + \delta_t \quad (7)$$

where  $\mu$  is the mean attention across frames and  $\delta_t = \|A_{FP16}^{(f+1)} - A_{FP16}^{(f)}\|_F$  is the baseline temporal difference. A sufficient condition for coherence preservation is:

$$\|\Delta QK^T\|_F \leq \delta_t \cdot \|\mu\|_2 \quad (8)$$

This theorem guides our adaptive quantization strategy: allocate higher precision (smaller  $\epsilon$ ) to attention heads with large temporal gradients.

## IV. ALGORITHMIC FRAMEWORK

### A. Adaptive Block-Wise Quantization

Standard block-wise quantization [?] divides tensors into fixed-size blocks, each with its own scale factor. For video diffusion, we observe that spatial and temporal dimensions have different sensitivity to quantization.

**Spatial sensitivity** arises from fine-grained textures and edges. **Temporal sensitivity** arises from motion continuity and object persistence.

Our **adaptive block-wise quantization** dynamically determines block size per layer:

$$\text{BlockSize}(l) = \underset{b \in \mathcal{B}}{\text{argmin}} \left( \text{Error}(l, b) + \lambda \cdot \frac{\text{Memory}(b)}{\text{Memory}_{\max}} \right) \quad (9)$$

where  $\mathcal{B} = \{1, 2, 4, 8, 16, 32, 64\}$  (block sizes along spatial dimension),  $\text{Error}(l, b)$  is measured on a calibration set, and  $\lambda$  balances accuracy and memory.

Algorithm ?? summarizes the adaptive selection procedure.

#### Algorithm 1 Adaptive Block-Wise Quantization

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**Require:** Layer  $l$ , calibration set  $D_{\text{cal}}$ , candidate blocks  $\mathcal{B}$ , trade-off  $\lambda$   
**Ensure:** Optimal block size  $b^*$   
1: **for** each  $b \in \mathcal{B}$  **do**  
2:   Quantize layer  $l$  with block size  $b$   
3:   Compute reconstruction error  $\text{err}[b]$   
4:   Compute memory ratio  $\text{mem}[b]$   
5:   Compute cost:  $\text{cost}[b] = \text{err}[b] + \lambda \cdot \text{mem}[b]$   
6: **end for**  
7:  $b^* = \underset{b}{\text{argmin}} \text{cost}[b]$   
8: **return**  $b^*$

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### B. Quantization-Aware Training for Cross-Modal Attention

Cross-modal attention between visual latents  $z_{\text{vis}}$  and audio features  $z_{\text{audio}}$  is critical for synchronization. We propose **learnable scale factors** per attention head:

$$\text{Attention}(Q, K, V) = \text{softmax} \left( \frac{QK^T}{\sqrt{d_k}} + \log(s_{\text{head}}) \right) V \quad (10)$$

where  $s_{\text{head}} \in \mathbb{R}^H$  are learnable scale factors initialized to 1.

The FP4 quantization simulation during training follows a straight-through estimator approach: during the forward pass, we quantize to FP4 and dequantize; during the backward pass, gradients bypass the quantization operation using the identity function. This enables the model to learn parameters that are robust to quantization error.

Algorithm ?? details the QAT procedure.

### C. Loss Functions

We optimize a combined loss:

$$\begin{aligned} \mathcal{L}_{\text{total}} = & \underbrace{\mathbb{E}_{t, \epsilon, x_0} \|\epsilon - \epsilon_\theta(x_t, t, c)\|_2^2}_{\text{Diffusion loss}} \\ & + \lambda_{\text{sync}} \underbrace{\text{SyncNet}(x_{\text{frames}}, c_{\text{audio}})}_{\text{Synchronization loss}} \\ & + \lambda_{\text{temp}} \underbrace{\|x_t - x_{t-1}\|_F^2}_{\text{Temporal smoothness}} \end{aligned} \quad (11)$$

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**Algorithm 2** Quantization-Aware Training for NVFP4-DiT
 

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**Require:** Pretrained BF16 model  $M$ , training data  $\mathcal{D}$ , learning rate  $\eta$   
**Ensure:** FP4-quantized model  $M_q$

```

1: Initialize  $M_q$  with weights from  $M$ 
2: Insert fake quantization ops (Algorithm ??) after each linear and attention
   layer
3: for epoch = 1 to  $E$  do
4:   for batch in  $\mathcal{D}$  do
5:     # Forward pass with simulated FP4
6:     pred =  $M_q$ (batch)
7:     loss =  $\mathcal{L}_{\text{diffusion}}$ (pred, batch) +  $\lambda_{\text{sync}} \mathcal{L}_{\text{sync}}$ (pred, batch)
8:     # Backward pass with straight-through estimator
9:     loss.backward()
10:    for each learnable scale factor  $s$  do
11:      Apply gradient scaling for stable updates
12:    end for
13:    optimizer.step()
14:  end for
15: end for
16: return  $M_q$ 

```

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where SyncNet [?] computes lip-sync error between generated frames and audio.

## V. SYSTEMS OPTIMIZATION

### A. FP4-Packed Triton Kernels

Naive FP4 implementation suffers from three main inefficiencies: (1) memory overhead from unpacking FP4 to FP16 for computation, (2) low tensor core utilization for non-standard data types, and (3) poor cache locality for block-wise scales.

Our Triton kernels address these challenges through three key innovations. First, we pack two FP4 values into a single byte, reducing memory bandwidth by  $4\times$  compared to FP16. Second, we perform on-the-fly dequantization directly within tensor core operations, avoiding intermediate FP16 expansion. Third, we optimize block-wise scale loading with vectorized memory accesses to maximize cache locality.

The kernel operates as follows: given input matrices stored in packed FP4 format with per-block scale factors, the kernel loads packed bytes, unpacks them to FP16 using the corresponding scales, performs matrix multiplication using Tensor Cores, and accumulates results in FP32. This design achieves:

- $2.1\times$  speedup vs. naive FP4 (unpack  $\rightarrow$  FP16  $\rightarrow$  compute)
- $3.2\times$  speedup vs. FP16 baseline
- $4\times$  memory reduction for weights and activations

### B. vLLM Integration for Video Diffusion

We adapt vLLM’s continuous batching for video generation. The key challenge is that video diffusion has variable-length generation (different numbers of frames per request) and variable compute per step (attention complexity scales quadratically with frame count  $F$ ).

Our solution implements three components:

**Frame-level paged attention:** The key-value cache is partitioned into fixed-size pages at the frame granularity. Each page stores attention states for a contiguous block of frames. This enables efficient memory sharing across requests and reduces fragmentation.

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**Algorithm 3** Continuous Batching for Video Diffusion
 

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**Require:** Request queue  $\mathcal{Q}$ , GPU memory budget  $M$   
**Ensure:** Generated videos

```

1: Initialize frame_kv_cache = {}
2: while  $\mathcal{Q}$  not empty do
3:   # Dynamic batching
4:   batch = []
5:   for each request  $req$  in  $\mathcal{Q}$  do
6:     if  $\text{memory\_estimate}(req) + \text{current\_memory} < M$  then
7:       batch.append( $req$ )
8:     end if
9:   end for
10:  # Prepare batch tensors (pad to max frames in batch)
11:  batch_padded = pad_to_max_frames(batch)
12:  # Run one diffusion denoising step
13:  pred = model(batch_padded, frame_kv_cache)
14:  # Update cache asynchronously
15:  frame_kv_cache.update(pred)
16:  # Remove finished requests from queue
17:   $\mathcal{Q} = \{req \in \mathcal{Q} \mid \neg req.done\}$ 
18: end while

```

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**Dynamic frame batching:** Requests are grouped by their remaining frame count. Batches are formed dynamically before each diffusion step, selecting requests with similar remaining frames to minimize padding overhead.

**Speculative frame decoding:** Multiple future frames are generated in parallel using a lightweight predictor, then verified by the full model with rejection sampling. This trades marginal compute for significant latency reduction when predictions are accurate.

Algorithm ?? shows our inference serving scheduler.

## VI. EXPERIMENTS

### A. Experimental Setup

#### Datasets:

TABLE I  
DATASET STATISTICS

| Dataset    | Type        | Size       | Resolution |
|------------|-------------|------------|------------|
| WebVid-10M | Text-Video  | 10M clips  | 336x336    |
| VGGSound   | Audio-Video | 200k clips | 224x224    |
| UCF-101    | Action      | 13k videos | 320x240    |

#### Models:

- DiT-S (small), DiT-B (base), DiT-L (large) [?]
- Stable Video Diffusion (SVD) [?]
- ModelScope-T2V [?]

#### Hardware:

- $8\times$  NVIDIA H100 (80GB) for training
- $1\times$  H100 for inference benchmarks

#### Baselines:

- BF16 (full precision)
- FP8 (E4M3) with QAT [?]
- INT8 with SmoothQuant [?]
- FP4 with static quantization

#### Metrics:

- **FVD** (Fréchet Video Distance): lower is better
- **SyncNet** accuracy: lip-audio sync (0-1)
- **CLIP-T** score: text-video alignment
- **LIPIPS**: temporal consistency (lower is better)
- Throughput (videos/second), Latency (ms/frame), Memory (GB), Energy (J/video)

## B. Main Results

TABLE II  
QUALITY COMPARISON ON WEBVID-10M TEST SET

| Method             | Precision | FVD   | SyncNet | CLIP-T | LPIPS |
|--------------------|-----------|-------|---------|--------|-------|
| DiT-B (baseline)   | BF16      | 98.3  | 0.892   | 0.312  | 0.087 |
| DiT-B + FP8        | FP8       | 100.1 | 0.885   | 0.308  | 0.089 |
| DiT-B + INT8       | INT8      | 105.7 | 0.871   | 0.301  | 0.094 |
| DiT-B + static FP4 | FP4       | 187.3 | 0.723   | 0.241  | 0.156 |
| NVFP4-DiT (ours)   | FP4       | 99.2  | 0.886   | 0.309  | 0.088 |

Key observations:

- NVFP4-DiT achieves 0.9% FVD degradation vs. BF16 (99.2 vs. 98.3)
- SyncNet accuracy drops only 0.7% (0.886 vs. 0.892)
- Static FP4 collapses completely (FVD 187.3)

TABLE III  
QUALITY ACROSS MODEL ARCHITECTURES

| Model | Precision | FVD   | SyncNet | Memory (GB) |
|-------|-----------|-------|---------|-------------|
| DiT-S | BF16      | 112.4 | 0.854   | 14.2        |
| DiT-S | NVFP4-DiT | 113.8 | 0.849   | 3.2         |
| DiT-B | BF16      | 98.3  | 0.892   | 28.5        |
| DiT-B | NVFP4-DiT | 99.2  | 0.886   | 6.8         |
| DiT-L | BF16      | 87.6  | 0.917   | 56.3        |
| DiT-L | NVFP4-DiT | 88.9  | 0.912   | 13.5        |
| SVD   | BF16      | 82.1  | 0.903   | 42.1        |
| SVD   | NVFP4-DiT | 83.4  | 0.898   | 10.1        |

Memory reduction: 76.3% on average.

TABLE IV  
INFERENCE PERFORMANCE (BATCH SIZE=4, 32 FRAMES)

| Method           | Throughput (videos/s) | Latency (ms/frame) | Memory (GB) | Energy (J/video) |
|------------------|-----------------------|--------------------|-------------|------------------|
| BF16 baseline    | 0.31                  | 101.2              | 28.5        | 892              |
| FP8              | 0.67                  | 46.8               | 14.3        | 412              |
| INT8             | 0.72                  | 43.6               | 12.1        | 384              |
| NVFP4-DiT (ours) | 1.05                  | 29.8               | 6.8         | 285              |

Improvements vs. BF16:

- Throughput:  $3.4\times$  (1.05 vs. 0.31 vid/s)
- Latency:  $3.4\times$  reduction (29.8 vs. 101.2 ms/frame)
- Memory: 76.3% reduction (6.8 vs. 28.5 GB)
- Energy: 68% reduction (285 vs. 892 J/video)

## C. Audio-Visual Synchronization Results

TABLE V  
SYNCNET ACCURACY ON VGG SOUND UNDER NOISE CONDITIONS

| Condition                     | BF16  | NVFP4-DiT | Difference |
|-------------------------------|-------|-----------|------------|
| Clean audio                   | 0.892 | 0.886     | -0.006     |
| + 10dB noise                  | 0.845 | 0.839     | -0.006     |
| + 20dB noise                  | 0.781 | 0.774     | -0.007     |
| Cross-modal (different audio) | 0.023 | 0.021     | -0.002     |

Finding: NVFP4-DiT preserves synchronization accuracy across noise conditions, with degradation consistently below 0.7%.

## D. Training Efficiency

TABLE VI  
TRAINING COST COMPARISON (DiT-B ON WEBVID-10M)

| Precision | GPU hours | Energy (MWh) | Cost (\$) |
|-----------|-----------|--------------|-----------|
| BF16      | 4,800     | 28.8         | 9,600     |
| FP8       | 2,400     | 14.4         | 4,800     |
| NVFP4-DiT | 1,600     | 9.6          | 3,200     |

Reduction: 66.7% GPU hours, 66.7% energy, 66.7% cost. These gains suggest that the proposed low-precision training pipeline can reduce end-to-end computational expense while preserving the quality trends reported in the preceding results. In practical deployment settings, this reduction can translate to shorter retraining cycles, lower serving cost per generated clip, and improved accessibility for iterative model development.

## VII. ABLATION STUDIES

### A. Block Size Analysis

Table ?? summarizes how quantization block size affects generation quality.

TABLE VII  
EFFECT OF BLOCK SIZE ON FVD (DiT-B, WEBVID-10M)

| Block Size        | FVD         | SyncNet      |
|-------------------|-------------|--------------|
| 1x1 (per-element) | 98.7        | 0.888        |
| 4x4               | 98.9        | 0.887        |
| 8x8               | 99.2        | 0.886        |
| 16x16             | 100.1       | 0.882        |
| 32x32             | 104.3       | 0.871        |
| Adaptive (ours)   | <b>98.9</b> | <b>0.888</b> |

### B. QAT Ablation

Table ?? isolates the contribution of each quantization-aware training component.

TABLE VIII  
IMPACT OF QAT COMPONENTS

| Configuration               | FVD         |
|-----------------------------|-------------|
| No QAT (PTQ only)           | 187.3       |
| + Standard QAT              | 112.4       |
| + Learnable scales          | 102.1       |
| + Cross-modal attention QAT | 99.8        |
| + Temporal smoothness loss  | <b>99.2</b> |

### C. Kernel Performance

Table ?? reports the per-layer runtime of our packed FP4 kernels against the baselines.

TABLE IX  
KERNEL MICROBENCHMARKS (DiT-B ATTENTION LAYER)

| Kernel                        | Time (us) |
|-------------------------------|-----------|
| FP16 baseline                 | 245       |
| Naive FP4 (unpack to compute) | 187       |
| FP4 Triton (no packing)       | 132       |
| FP4 Triton (packed)           | 89        |
| FP4 FlashAttention (ours)     | <b>67</b> |

## VIII. DISCUSSION

### A. Why Does NVFP4-DiT Succeed Where Static FP4 Fails?

Static FP4 collapses because:

- 1) **Extreme outliers** in attention logits cause saturation (clip to  $\pm 6$  in scaled domain)
- 2) **Temporal variations** exceed block-wise scale range
- 3) **Cross-modal gradients** are mis-scaled for low precision

Our adaptive block-wise quantization dynamically adjusts block size per layer, while learnable scales in cross-attention optimize the dynamic range for audio-visual fusion. The temporal smoothness loss regularizes the model to produce coherence-respecting outputs.

### B. Limitations

- 1) **Training complexity:** QAT adds 20% overhead vs. BF16 training (but total training is 66% shorter due to FP4 acceleration)
- 2) **Hardware requirement:** NVFP4 tensor cores only available on H100/B200 (not Ada/Ampere)
- 3) **Long video generation (over 128 frames):** Accumulated quantization error may degrade quality; we are exploring hierarchical quantization strategies
- 4) **Theoretical bounds are worst-case:** In practice, degradation is much smaller (under 1%)

### C. Societal Impact

**Positive:** Democratizes video generation by reducing compute cost; enables on-device creative tools; lowers carbon footprint of generative AI.

**Negative:** Could lower the barrier for deepfake generation; responsible deployment should therefore include dataset governance, provenance mechanisms, and downstream content safeguards.

## IX. CONCLUSION

This paper presented NVFP4-DiT, a framework for studying 4-bit training and inference for audio-guided video diffusion transformers. We developed a theoretical perspective on FP4 diffusion, introduced adaptive block-wise quantization and QAT for cross-modal attention, and described implementation choices for low-precision kernels and serving. The experimental results indicate that carefully designed 4-bit training can substantially reduce memory and computation while retaining competitive multimodal generation quality. These findings support further investigation into reliable low-precision training for video generation systems.

Additional implementation details, checkpoints, and qualitative examples are provided in the supplementary material.

### CODE AVAILABILITY

Code, kernels, and related artifacts will be released upon publication.

### CONFLICTS OF INTEREST

The authors declare no conflicts of interest.

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